A Retroactive Evaluation of a Games-Based Health Education Curriculum

Grassroots Health

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08.06.2026

Abstract

Washington, DC faces a unique set of health challenges among the large youth population. Youth in DC are facing disproportionate rates of childhood obesity, STDs/STIs, as well as struggling with low mental health scores and rates of flourishing.

Grassroots Health (GRH) is a 501(c)(3) nonprofit organization based in Washington, DC, that trains NCAA athletes to be health educators in public middle schools. The goal of Grassroots Health is to use the platform of athletes and a near-peer-based education model to minimize stigma around nutritional, sexual, and mental health. Through a three-year games-based educational curriculum that is scaffolded by health and P.E. classes, Grassroots Health's vision is to change the "culture of health" among young people.

In the 2025-2026 school year, GRH worked with more than 1,700 students across 14 middle schools. Across these schools, 577 sixth graders received nutrition programming, 599 seventh graders received sexual health programming, and 542 eighth graders received mental health programming. A quasi-experimental research project used surveys that students completed at the beginning and end of their semester-long programming. This pre/post survey dataset was analyzed as part of an internal program evaluation project to assess curriculum efficacy.

The State of Adolescent Public Health in the District of Columbia

According to the 2023-2024 National Survey of Children's Health (NSCH), 30% of children ages 6-17 in DC are classified as overweight or obese on the BMI scale (*State of Childhood Obesity* 2025). Of the 30%, 18% of the population is specifically obese, ranking D.C. ninth among the United States' overall childhood obesity rate for 2023-2024 (*State of Childhood Obesity* 2025). Children who live in Wards 5, 7, and 8 of DC maintain childhood obesity rates that surpass all states and the national average, likely due to a myriad of reasons, including: food insecurity, low income, and large minority populations (*District of Columbia—2018—III.E.3. State Action Plan*). A 2022 study found that children who possessed a social network that encouraged healthy eating, either through peers or family members, engaged in more healthy eating behaviors and had a better quality diet than those without a support network (Berge et al 2022). If these networks are engaged more through community outreach or focus groups, intervention teams may gain a better understanding of what specific populations need.

As of 2024, approximately 47% of children in DC are Black non-Hispanic, 23% are white, 20% are Hispanic, and 10% fall into other categories (*NSCH Interactive Data Query (2022—Present)*). Racial discrimination is a risk factor for child and adolescent obesity, independent of socioeconomic status or food insecurity. A 2023 study found that children with reported racial discrimination experiences at their baseline survey were associated with higher adiposity at follow-up one year later (Cuevas 2023). Furthermore, children who experienced obesity also reported the highest mean overall racial discrimination scores (Cuevas 2023).Identifying systematic determinants of malnutrition is important whenever serving primarily nonwhite communities, such as DC. Whenever curriculums are tailored to cultural and demographic contexts of the population they serve, they have been found to be more successful (Murimi et. al 2018).

In addition to nutritional health, sexual health knowledge, literacy, and confidence are important areas of improvement for middle schoolers in Washington, DC. According to the 2023-2024 Youth Risk Behavior Survey (YRBS), 7% of middle school students in DC had had sex (CDC 2025). Of that number, only 47% of middle schoolers used condoms during sexual intercourse (CDC 2025). Additionally, 15% of DC middle schoolers reported that they had experienced physical dating violence (CDC 2025). Young people in DC are engaging in sexual behaviors, indicating that there is a meaningful opportunity to alter existing sex education and encourage healthier and safer practices. Health education has been widely shown to decrease sexual debut and increase positive protective measures, such as condom use, testing services, and pregnancy prevention. A study of participants ages 15-19 years old found that those who received sex education were 1.42 times more likely to not have had sexual intercourse than those who had not received sex education (Mueller 2008). Males who received sex education were also 2.77 times more likely to use contraception the first time they had sexual intercourse, compared to those who did not receive sex education (Mueller 2008).

Teenagers in DC face STI rates that are more than double the national average. Among 13-17 year olds in DC, there were 1,117 new cases of syphilis, gonorrhea, and chlamydia in 2024 (*Selected Indicators for District of Columbia | KIDS COUNT Data Center* 2024). This is equivalent to 23.28% of all 13-17 year olds in DC. In the United States as a whole, approximately 11% of young people ages 15-24 are diagnosed with an STI annually. These heightened rates compared to national averages signify a pressing need for more culturally competent and tailored sexual health education for DC youth. Comprehensive sexuality education, or CSE, has been found to decrease pregnancy rates and safe-sex behaviors among populations of adolescent females in the United States (Bordogna 2022). While the rates of STIs in DC are extremely high, CSE has a strong basis for decreasing sexual activity and risk.

A 2022 systemic review of literature found moderate effect sizes of school-based interventions targeting emotional regulation among middle schoolers (Pedrini 2022). Middle schoolers in DC report high rates of mental health struggles in the 2023 YRBS, 16% said that their mental health is most of the time or always not good (CDC 2025). The National Survey of Children's Health measures child wellbeing and mental health based on the number of 'flourishing items' that children are meeting, showing interest and curiosity in learning new things, working to finish tasks that they start, and staying calm and in control when something is challenging. Approximately 40% of children in DC are only meeting 0-2 of all 3 flourishing items (*NSCH Interactive Data Query 2022—Present*). Additionally, 34% of high school students report feeling sad or hopeless (*NSCH Interactive Data Query 2022—Present*). Prevention-based and resilience-enhancing programming in middle school can help alleviate these kinds of mental health responses later in adolescence, as children learn the tools needed for recognizing stressors (McGovern 2022).

The Grassroots Health Theoretical Framework

Social Ecological Model & Social Learning

Grassroots Health uses a games- and peer-based model of education in order to reduce stigma , encourage students to value their personal health, and have the information necessary to make informed decisions and self-advocate. The Social Ecological Model (SEM) theorizes that individual behaviors and health outcomes are heavily influenced by culture, one's environment, and the framework of their society.

Because of this, public health professionals view social-ecological based interventions as effective solutions whenever they are adapted to fit the specific needs of the community (Ericksson 2025).

GRH employs the SEM and Social Learning theories to tailor an intervention that best serves middle schoolers in DC. While familial influence is the most influential in the first years of a child's life, it is surpassed by peer relationships during middle school years (Ekubagawargies 2025). Additionally, the ability to implement new curricula and shift culture in schools makes them a more productive setting for intervention, as opposed to in-home. School-based interventions allow the entire target population to participate, and not just the subset of youth and families who opt-in . By training NCAA athletes to be mentors and health educators, GRH creates role models and peers with the potential to have a significant impact on a student's learning experience.

Peer-assisted learning, or PAL, is the practice of pairing lower-achieving students with higher-achieving peers in educational settings (Helseth 2017). Studies have shown that the impact of PAL interventions in Lower-Middle Income Countries is especially significant (Ekubagewargies 2025). This logic can be applied to the GRH target population, 19% of families in DC fell below the Federal Poverty Line (FPL) between 2023 and 2024, and 12% of families were between 100% and 199% of the FPL (*NSCH Interactive Data Query 2022—Present*). The GRH curriculum has implemented PAL between students in Title 1 schools (lowest economic status) and NCAA athletes, by pairing students with older and high-achieving mentors, schools are more likely to see vast improvements in students' attitudes towards nutritional, sexual, and mental health.

PAL is impactful due to its provision of role models that influence child development, learning, and self-efficacy. One 2002 study with an average sample age of 14.5 years old showed that those who had role models earned higher grades and better self-esteem (Yancey 2002). Additionally, 72% of children in the sample chose role models who shared the same ethnicity as themselves (Yancey 2002). By recruiting and training athletes from colleges and universities in the DC area, GRH provides role models to students who have shared identities and similar life experiences. *Table 1* shows demographics of students served in the 2025-2026 school year compared to demographics of the athlete coaches who worked with them.

Table 1. Demographics of students vs. athlete mentors.

	Students (%)	Athlete Mentors (%)
Gender		
Man	48.05	21.46
Woman	46.03	78.45
Transgender	1.59	0.09
Ethnicity		
Asian	0.77	0.75
Black or African American	68.89	63.53
White or European American	1.52	15.04
Hispanic	21.21	2.07
Mixed	2.91	6.58

Other	4.7	12.03

Whole School, Whole Community, Whole Child

The Grassroots Health curriculum is further rooted in the Whole School, Whole Community, Whole Child (WSCC) framework. The WSCC is focused on the culture of health and the importance of coordinated care between all realms of a student's life. Under the WSCC framework, child health is seen as holistic and determined by the health, culture, and beliefs of their families, teachers, and communities (Jarpe-Ratner 2025). In addition to in-school programming directly with students, GRH also runs Grassroots Fam and Grassroots Connect programs, both with the goal of providing support to caregivers and community members who interact with students. Grassroots Fam is a two-session workshop provided for parents and caregivers at GRH's partner schools, with the goal of improving health communication and literacy surrounding the topics students cover in programming. Grassroots Connect is a biannual health fair that engages students with local providers in their community, offering social and clinical services.

Kinesthetic Learning

Finally, the Grassroots Health curriculum is based on kinesthetic learning methods. Kinesthetic learning is an education model based on physical engagement among students in their lessons, and can look like role playing, simulations, physical movement, etc. (Oladele 2024). Kinesthetic learning engages students' touch, spatial awareness, and visual senses. This engagement encourages the retainment and retrieval of information, and can also help develop a student's critical thinking and problem-solving skills (Oladele 2024). Because the Grassroots Health curriculum heavily emphasizes self efficacy and decision-making, a learning model that further develops these skills is highly productive. A 2014 study conducted among eighth graders in Baltimore, MD found that whenever implementing theatrical role-playing recreations of material learned in class, students in the experimental group were more likely to engage in collaborative work, be confident in their abilities to answer questions, and be proud of their academic achievements (Gonzales 2014). Through a mixture of physical activities, as well as discussions led by coaches and visual aids, the Grassroots Health curriculum encourages concept retention, decision-making, and reaches more learners.

The Value of Prevention

A large component of the GRH framework focuses on improving self-efficacy among young people. Students learn to make choices for themselves regarding what kinds of foods will fuel their bodies best, how to speak up for themselves when it comes to sex and consent, and how to recognize stressors and ask for help with their mental health. Studies have found that an increase in parent-perceived self-efficacy among children translates to a higher success rate in navigating the challenges of transitioning from paediatric to adult healthcare (Kwok 2025). Additionally, self-efficacy rates correlated with the highest rates of general adherence to healthy behaviors, doctors appointment attendance, and engagement in personal healthcare needs (Kwok 2025).

GRH's prevention-focused intervention addresses the burden of non-communicable disease among children and adolescents. Many risk factors, such as unsafe sex and mental health challenges, are largely preventable, and this prevention begins in the early stages of adolescence (Catalano 2012). While it is

difficult to track and sustain longitudinal studies that would provide concrete data regarding the correlation between health education in middle school and health behaviors in early to late adulthood, the need for education still remains. Prevention work not only provides foundational knowledge of risk factors and resources available to students, but also teaches them how to advocate for themselves and use those resources whenever they are needed. The GRH curriculum also increases awareness of both healthy and unhealthy behaviors, and encourages positive peer interactions that shift the culture of health.

The Grassroots Health Curriculum

Games-Based Learning

Grassroots Health relies on a games-based learning method that incorporates physical activity while teaching health curricula. Each student will receive 8-10 sessions, incorporating more than 15 total activities per intervention. Each student will complete a nutritional health program in sixth grade, a sexual health program in seventh grade, and a mental health program in eighth grade. The GRH curriculum revolves around the multiple avenues of a student's life that influence health decisions, as shown by the WSCC model and SEM. Programming engages with students, school administrators, parents, and the community in order to build a framework that shifts the culture of health. The kinesthetic learning model of Grassroots Health helps students build decision-making skills that will aid them when it comes to their personal health, as well as encourages retention of important information. A curriculum rooted in prevention gives students lifelong skills, rather than mitigating a one-time problem. Through games such as *Food Group Frenzy*, *Hormone Monster, Crossing the Line*, and *Stress Football*, students participate in both engaging movement and impactful conversations with athlete mentors. These are just 4 examples of more than 30 activities.

Food Group Frenzy is a relay race game that reviews the three major nutrient groups, identifies foods that contain each nutrient, and explains how each nutrient keeps the body healthy. A relay race is set up with hurdles, a speed ladder, and a sprint to the final cone. In the first round, students race through the three stages of the relay without any obstacles, reviewing the major nutrient groups. In the second round, they complete the hurdle section with an added challenge of a resistance band, demonstrating how a lack of protein can challenge their strength. In the third round, they complete the speed ladder with the added challenge of a blindfold, showing the impact of a lack of vitamins and minerals on their ability to concentrate. In the fourth and final round, instead of sprinting to the cone, students will bear crawl or lunge, demonstrating how a lack of carbohydrates will challenge their energy levels. At the end of the game, students and athletes discuss the ways that different foods allow our bodies to accomplish different tasks, and how students' bodies had a more difficult time getting through strength, concentration, and energy sections of the race based on which 'nutrient' they were lacking, thus emphasizing the need for a balanced diet.

Hormone Monster is one of the introductory games in the Sexual Health curriculum, and represents the ways hormones impact students' bodies as they age. In this version of 4-way dodgeball, teams follow normal dodgeball rules. The first round is played with 1 ball per team that represents hormones, each round represents a new adolescent age, starting with age 8 to see how hormones change as you age. The second round is played with 2 balls per team, and the third round with 3. Students discuss how the game

got overwhelming, especially in the third round, and how this relates to how they can feel as they get older and their bodies produce more hormones.

Crossing the Line is a partner game aimed at helping students learn about consent, pressure, and setting boundaries. In the first round, students are not allowed to speak and answer questions such as "Do you listen to music more than you read?" by standing on one side of a line for yes, and one side for no. This round represents individual decision-making. In a second round, partners run a 3-legged race without talking to each other or communicating. In a third round, partners run a 3-legged race again, but are allowed 30 seconds to communicate. The fourth round consists of harder questions that partners are allowed to discuss before crossing the line, such as "Are you comfortable with your partner holding your hand at school?". Finally, partners run a three-legged race with blindfolds on to represent consent while under the influence - they are able to communicate, but are unable to see the course. Students discuss how making choices together was easier when they were able to communicate, and how being under the influence renders communication unclear and consent can only happen when both partners are sober.

Stress Football is played with the goal of students understanding what acute, chronic, and compounding stress is. In *Stress Football*, students are organized into two lines, quarterbacks and receivers. In the first round, the first receiver in line will run to the end zone and catch the football from the quarterback. The receiver runs the football back to the next quarterback in line, and switches lines. Quarterbacks become receivers, and vice versa, until everyone has caught and thrown. In the second round, a student volunteers to be a defender in the end zone. The game is played the same way, except the defender can anticipate and intercept the quarterback's pass. After the second round, coaches and students discuss what acute stress is, and how the defender represents the normal and short-term stress response. Defenders can be seen and avoided, but also make you a better player. Stress isn't "all bad" but it is important to understand and identify daily, acute stressors. The same task as the first round was more difficult whenever a stressor was added to the equation, but manageable. In a third round, receivers wear a resistance band that a coach pulls back on as they run to the end zone, making it challenging to get all the way to the end zone and catch the pass. After round three, students and coaches discuss how the resistance band represents chronic stress, it was harder to run, it pulled students back the entire time, and they could not avoid it. Chronic stressors include things that happen over a sustained amount of time, like struggles with family, conflict, grief, community safety, etc. In a fourth round, receivers run with resistance bands, and there are now two defenders in the end zone. Afterwards, coaches explain that this represents the compounding factors of both kinds of stress. There were two defenders this round because chronic stress over time can make acute stress even more difficult to handle.

Students are able to learn about balanced meals, safe sex, and what impacts their mental health in ways that are kinesthetically, visually, and auditorily engaging. By learning in multiple ways that are fun and memorable, the physical representation of these topics can be recalled and built on week after week. The atmosphere of a games-based learning environment also decreases stigma and increases confidence in asking questions, as it does not feel as intense as a classroom or lecture environment.

Pre- and Post-Intervention Surveys

The effectiveness of the GRH curriculum is evaluated using pre- and post-intervention surveys that students take at the beginning and end of the semester. GRH also shares survey results with their partner schools so that faculty and staff can understand what conversations GRH is starting, where students are still struggling, and where their attention can increase support. *Table X* describes each component of pre- and post-intervention surveys, and the theoretical basis for the constructs included.

	Test Items	Content	Theoretical Basis	Existing Literature
Nutrition				
Knowledge	6 multiple choice questions	The job of proteins, vitamins and minerals, and carbohydrates, and what foods contain each macronutrient	Health Literacy	Adapted from the *General Nutrition Knowledge Questionnaire (GNKQ)* (Parmenter and Wardle 1999)
Outcome Expectations	4 Likert Scale questions, from *Strongly Disagree* to *Strongly Agree*	Beliefs surrounding the impacts of healthy eating	Social Cognitive Theory	Adapted from the *Healthy Eating and Weight Self-Efficacy (HEWSE) Scale* (Wilson-Barlow 2014)
Outcome Expectations Values	3 Likert Scale questions, from *Strongly Disagree* to *Strongly Agree*	Personal values assigned to healthy eating habits	Social Cognitive Theory	Adapted from the *Healthy Eating and Weight Self-Efficacy (HEWSE) Scale* (Wilson-Barlow 2014)
Behaviors	5 Likert Scale questions, from *Never* to *Always*	Frequency of behaviors such as choosing to drink water or eat a healthy meal	Social Cognitive Theory	Adapted from the *Adolescent Healthy Eating Questionnaire* (Dewar 2025)
Sexual Health				
Knowledge	11 T/F/Unsure questions	How HIV/STIs spread, proper forms of protection, and effective ways	Health Literacy	Adapted from the *Brief HIV Knowledge Questionnaire*

		of avoiding unplanned pregnancy		(HIV-KQ-18) (Carey and Schroder 2008)
Self Efficacy	5 Likert Scale questions, from *I definitely cannot do this* to *I definitely can do this*	Advocate for oneself regarding sex and general consent with a partner	Social Cognitive Theory	Partially adapted from the *Condom Use Self-Efficacy Scale (CUSES)* (Brafford 1991)
Attitudes + Behaviors	9 Likert Scale questions, from *Strongly Disagree* to *Strongly Agree*	Willingness to ask questions about sex, have conversations regarding sex, and lacking fear of judgement	Health Belief Model	Based on Ajzen's *Theory of Planned Behavior (TPB)* (Ajzen 1991)
Peer Communications	7 Likert Scale questions, from *None/Unsure* to *A lot*	Frequency of discussion among peers regarding prevention, HIV/STIs, and consent	Social Cognitive Theory + Social Norms Theory, WSCC	Developed from measures included in the *Youth Risk Behavior Survey (YRBS)*
Parent Communications	7 Likert Scale questions, from *None/Unsure* to *A lot*	Frequency of discussion among parents/guardians regarding prevention, HIV/STIs, and consent	Social Cognitive Theory + Social Norms Theory, WSCC	Developed from measures included in the *Youth Risk Behavior Survey (YRBS)*
Mental Health				
Expectations	8 Likert Scale questions, from *Strongly Disagree* to *Strongly Agree*	Fear of judgement, reasons for seeking help, desire to be open regarding mental health struggles	Health Belief Model	Adapted from the *Mental Help Seeking Attitudes Scale (MHSAS)* (Hammer JH et al. 2018)
Values	9 Likert Scale questions, from *Not at all important* to *Extremely important*	Asking for help and dealing with negative emotions, understanding where emotions come from, and having healthy boundaries and	Social Emotional Learning	Adapted from the *Social Emotional Assets and Resilience Scales (SEARS)* (Klages 2023)

relationships

Self Efficacy	7 Likert Scale questions, from *Strongly Disagree* to *Strongly Agree*	Ability to recognize and manage stress, calm down, step away from harmful situations, and ask for help	Social Cognitive Theory	Adapted from the *Self-Efficacy Questionnaire for Children (SEQ-C)* (Muris 2001)*)*
Self Control	6 Likert Scale questions, from *Not at all true* to *Really true*	Ability to recognize frustration, control temper, and think through actions	Self-Regulation Theory, Social Emotional Learning	Adapted from the *Brief Self-Control Scale* (Manapat 2021) and the *Adolescent Self-Regulatory Inventory (ASRI)* (Moilanen 2007)
Well Being	8 Likert Scale questions, from *Strongly Disagree* to *Strongly Agree*	Feelings of hopefulness and being useful, clear thoughts, self confidence, and close relationships to friends and family	Social Emotional Learning	Adapted from the *Warwick-Edinburgh Mental Well-being Scale (WEMWBS)* (Tennant 2007)

Data Management and Results

Methods

Pre-intervention data from Fall 2025 and Spring 2026 were imported from Google Forms into Microsoft Excel for cleaning. Survey responses were reviewed for completion and quality. This included removing any surveys with responses that seemed to have been done by an inattentive respondent, such as those with the same answer for every question. Student names were additionally checked, and any duplicate responses were removed from the dataset. After the data was cleaned, it was merged into a singular Microsoft Excel file encapsulating pre-intervention survey responses from the 2025-2026 school year. The same process was conducted for all post-intervention data from Fall 2025 and Spring 2026 as well.

Data was then coded, with all responses converted to binary variables. For multiple choice or true/false questions, correct responses received a value of 1, and incorrect or “Unsure” responses received a value of 0. Survey responses to questions that asked students to rank their level of agreement, confidence, behavior frequency, etc. on a Likert Scale received the number that coordinated with their response, from 1-5. For example, a response of “Unsure” or “Strongly Disagree” received a 1, and a response of “Strongly Agree” or “All the time” received a 5. Individual scores for each participant were totaled among each construct, shifting the Likert Scale to be dependent on the number of questions asked in each section.

A standard participant name was created for each survey respondent in Microsoft Excel by combining first and last names, converting all characters to uppercase, and removing any non-alphabetical characters from the columns. This was done in order to avoid as many formatting discrepancies as possible that would lead to matching issues. Records that were not matched were reviewed manually, and discrepancies caused by nicknames, misspellings, etc. were matched by verifying birth dates and school attended. This process resulted in 392 matched Nutrition surveys, 486 matched Sexual Health surveys, and 372 matched Mental Health surveys.

Results

Cleaned, coded, and matched data were imported into STATA in order to conduct paired-sample t-tests on all constructs of each curriculum. The paired-sample t-tests allow GRH to determine if there is a statistically significant difference between pre- and post-intervention survey scores. The results can be found in *Table 2*, *Table 3*, and *Table 4*. After initial data analysis, racial demographics among students and gender identity were also numerically coded. The data was analyzed to determine differences in score improvement across demographics. Full statistical output regarding racial demographics can be found in Appendix B, Table B1 - Table B14. The two most common demographics in the student population are Black or African American and Hispanic/Latino, and are the demographics examined the closest in analysis. Full statistical output regarding gender identity can be found in Appendix C, Table C1 - Table C14 . In the survey collection, students had the choice between Girl/Woman, Boy/Man, or Nonbinary. Stratification by gender identity did not produce any statistically significant results in the Nonbinary categorization. Of the three curricula, Sexual Health was the only construct that showed significant differences between gender identities. A $p < 0.05$ result is generally considered statistically significant, and indicates that there is less than a 5% chance that the results observed in the data occurred purely by chance.

Table 2. Mean correct knowledge and Likert score sums from Nutrition surveys of the 2025-2026 school year.

	Possible Scoring Range	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Knowledge	0-6	2.91 (2.75, 3.07)	4.09 (3.91, 4.27)	1.18	$p < .001$
Outcome Expectations	5-20	11.77 (11.52, 12.01)	15.70 (15.38, 16.03)	3.94	$p < .001$
Outcome Expectation Values	5-15	11.77 (11.52, 12.01)	11.75 (11.48, 12.01)	-0.02	0.89
Behaviors0.140.	5-25	15.53 (15.20, 15.85)	15.39 (15.03, 15.75)	0.14	0.47

Table 3. Mean correct knowledge and Likert score sums from Sexual Health surveys of the 2025-2026 school year.

	Possible Scoring Range	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Knowledge	0-11	3.65 (3.40, 3.89)	6.61 (6.35, 6.87)	2.96	$p < 0.001$
Self Efficacy	5-25	18.68 (18.14, 19.22)	20.21 (19.71, 20.71)	1.53	$p < 0.001$
Attitudes + Beliefs	9-45	24.73 (24.16, 25.30)	31.05 (30.41, 31.69)	6.31	$p < 0.001$
Peer Communications	7-28	12.76 (12.21, 13.31)	16.57 (15.91, 17.23)	3.81	$p < 0.001$
Parent Communications	7-28	12.55 (11.93, 13.18)	15.42 (14.72, 16.12)	2.87	$p < 0.001$

Table 4. Mean Likert score sums from Mental Health surveys of the 2025-2026 school year.

	Possible Scoring Range	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Expectations	8-40	37.60 (36.78, 38.41)	38.67 (37.89, 39.44)	1.07	0.06
Values	9-45	22.68 (22.17, 23.20)	23.26 (22.72, 23.79)	0.57	0.15
Self Efficacy	7-35	25.77 (25.14, 26.40)	26.50 (25.91, 27.09)	0.73	0.11
Self Control	6-30	19.25 (18.64, 19.85)	20.53 (19.92, 21.14)	1.29	0.004
Well Being	8-40	29.76 (28.99, 30.53)	30.36 (29.63, 31.09)	0.60	0.28

Analysis

Nutritional Health

Nutritional knowledge improved significantly between the pre-survey and post-survey. Students got an average of 1.18 ($p < 0.001$) additional nutrition knowledge questions correct in the post-survey. This indicates an increase in understanding regarding basic nutrient knowledge with less than 0.1% chance of results occurring by chance. A question asking students which food primarily provides carbohydrates to

the body saw the greatest improvement from pre- to post-test; 27% of students answered correctly in the pre-survey and 53% answered correctly in the post-survey. The Black and African American student population showed less improvement in Nutrition Knowledge than average, increasing by 1.05 ($p < 0.001$) points. While a minimal difference, this could point to the need for more culturally aligned Nutrition education. Many sixth graders do not yet have a great deal of choice in their food selections. However, ensuring that they have a foundational knowledge of nutrients and well-rounded meals sets students up for making healthy choices in the future.

Nutritional outcome expectancies improved by 3.94 points from pre- to post-survey ($p < 0.001$). There was a meaningful shift in the way students understand the relationship between healthy eating and it's impacts. The values students placed in these expectancies slightly decreased by 0.02 points ($p = 0.89$), although the decrease was not statistically significant. The way students view the relationship between healthy eating and their personal health outcomes is not meaningfully shifting, and could require edits in curriculum. Additionally, there were no statistically significant improvement rates among the population whenever stratified by race. Students scored highly on the behavioral construct of the pre-survey and maintained their results through the post-survey. A high percentage of students reported healthy habits such as choosing to drink water over a sugary drink when thirsty and preparing fruits and veggies in a way that would be more enjoyable for themselves. Because GRH programming spans only 12 weeks, meaningful behavioral changes may be unlikely; however, the consistency of healthy behaviors is worth noting.

Sexual Health

Sexual health knowledge increased an average of 2.96 points after receiving programming ($p < 0.001$). A question regarding whether or not there is a cure for HIV saw a 36% increase in correct answers. This could lead to an increase in accurate information spread among peers and more informed decision making. Black or African American populations increased by 2.54 ($p < 0.001$) points, slightly lower than the population average. Knowledge scores among the Hispanic/Latino population increased by 3.94 ($p < 0.001$) points. This indicates that the Hispanic/Latino population is scoring almost one more question correctly than the entire population on average. Students who categorized themselves as Girl/Woman had a knowledge increase of 3.33 ($p < 0.001$) points, whereas those who categorized themselves as Boy/Man saw an increase of 2.62 ($p < 0.001$) points.

Self-efficacy scores improved an average of 1.53 points ($p < 0.001$). A significant increase in ability to decline uncomfortable situations was seen at the end of GRH programming. The Black or African American population Self-efficacy scores increased by 1.42 ($p = 0.002$), showing a minor decrease from the population average. There was no significant difference between the Self-Efficacy score increases in boys versus girls. Students' average Attitude and Behavior score improved by 6.31 points ($p < 0.001$). Among the Black or African American population, Attitude and Behavior increased by 6.89 ($p < 0.001$) points. Among the Hispanic/Latino population, points increased by 4.95 ($p < 0.001$). Overall, these changes show significant de-stigmatization regarding discussions of and questions regarding sex. However, they also indicate that the Hispanic/Latino population is being underserved by the curriculum in comparison to other demographics. There was no significant difference between the Attitude and Behavior score increases in boys versus girls.

A statistically significant improvement in the frequency of both peer and parent communications surrounding sex was seen by 3.81 ($p < 0.001$) and 2.87 points ($p < 0.001$), respectively. Among the Black or African American population, peer and parent communications improved by 3.32 ($p < 0.001$) and 2.94 ($p < 0.001$) points respectively. Among the Hispanic/Latino population, these scores increased by 5.47 ($p < 0.001$) and 2.86 ($p < 0.001$) points. Peer communications increased significantly more in the Hispanic/Latino population than all other demographics. Notably, peer communication scores increased by 4.59 ($p < 0.001$) versus 3.10 ($p < 0.001$) points in girls and boys respectively. Across the board, students are discussing topics such as condom use, ways to prevent pregnancy, and consent with their peers more often. Due to an increase in knowledge, they are sharing more accurate information when doing so. An increase in parent communication signifies the de-stigmatization of conversations surrounding sex at home, and can lead to easier conversations in the future as students' sexual activity increases and questions related to safe sex behaviors arise.

Mental Health

Mental health self-control improved with statistical significance by 1.29 ($p = 0.004$) points between the pre- and post-surveys. Self control among the Hispanic/Latino population specifically improved by 1.82 ($p = 0.035$) points. All mental health constructs improved, but rates of improvement were not statistically significant outside of self-control. Additionally, Self Control was the only construct that saw statistically significant changes when stratified by race.. Similar to Nutrition constructs, behaviors surrounding mental health can be difficult to change in a relatively short period of time. Grassroots Health' main goal in the mental health space is to improve self-confidence in mental health and help students understand and identify when challenges arise. As issues such as depression and anxiety are difficult to remedy in a 12-week program, increases in self-confidence and knowledge of stressors and emotions is a success. With this information, students will be better able to advocate for themselves, recognize unhealthy patterns, and ask for help if they need it.

Recommendations

Further research is needed regarding new ways to implement the Nutrition curriculum of Grassroots Health and see statistically significant improvements in the Outcome Expectancy Values and Behaviors constructs. Focus groups may be held with families served by Grassroots Health to understand the kinds of foods and cuisines kids are used to seeing at home, and edits to the curriculum may be made to include more culturally relatable dishes. Additionally, the curriculum is currently under-serving boys in comparison to girls in the Knowledge and Peer Communication constructs of Sexual Health. Specifically in the Sexual Health curriculum, Grassroots Health may improve efficacy by tailoring games based on gender identity or ensuring that conversations regarding sexual health are generalizable to all genders.

Methods of pre- and post-intervention evaluation may be altered in order to better suit sixth graders specifically, as well as increase matching numbers across all three curricula. The nutrition surveys had the highest rate of unmatched or unusable survey responses, with 577 students receiving programming and 392 surveys being matched. Their younger age range, combined with test exhaustion at the end of the year, could be leading to more incomplete/inattentive responses. A form of pre- and post-intervention discussion group or other oral method of data collection may be better suited for this age group. A kind of

Grassroots Health Student ID may also be considered, where students are assigned a unique number by the Evaluation Team and enter it in their pre- and post-intervention surveys. This would allow for a more streamlined matching process.

A longitudinal study examining behavior, self efficacy, and knowledge of students involved in Grassroots Health programming versus those not involved over the course of adolescence and young adulthood would be highly valuable, and produce concrete statistics regarding the efficacy of prevention work. Grassroots Health cannot retain contact information from students served. If Grassroots Health began working with a middle school that funneled to a specific high school, there would be opportunity to conduct a health behavior study stratified by those who received GRH programming in middle school versus those who did not. This would allow GRH to quantify the longitudinal impact of prevention work.

Conclusion

The Grassroots Health program serves the middle school population of Washington, DC, with the goal of shifting the culture of health and giving students the tools they need to advocate for themselves. Through games-based education and NCAA athlete coaches, GRH provides role models to students that decrease stigma around conversations related to nutrition, sexual health, and mental health. Through a quasi-experimental research design and data analysis of pre- and post-intervention surveys among students, it can be found that the GRH curriculum leads to statistically significant changes among students across all three grade levels.

Appendix A. Pre- and Post-Intervention Survey.

Nutrition Knowledge

- Which nutrient's main job is to build muscle? (Proteins / Carbohydrates / Vitamins and minerals / Unsure)
- Which nutrients help protect us from getting sick and stay focused? (Proteins / Carbohydrates / Vitamins and minerals / Unsure)
- Which nutrients help provide energy? (Proteins / Carbohydrates / Vitamins and minerals / Unsure)
- Which food primarily provides protein? (Spinach / Chicken / Whole wheat bread / Unsure)
- Which food primarily provides carbohydrates? (Spinach / Chicken / Whole wheat bread / Unsure)
- Which food primarily provides vitamins and minerals? (Spinach / Chicken / Whole wheat bread / Unsure)

Nutrition Expectations and Values

Response scale: Strongly Disagree – Disagree – Neither Agree nor Disagree – Agree – Strongly Agree

- Healthy eating can reduce my risk for illness and disease as I get older.
- Reducing my risk for illness and disease as I get older by eating a healthy diet is important to me.
- Healthy eating can give me more energy.
- Having energy throughout the day is important to me.
- Healthy eating can help me focus at school.
- Staying focused or improving my focus at school is important to me.
- I am confident that if I eat a balanced diet, I can improve my health.

Nutrition Behaviors

Response scale: Never – Rarely – Sometimes – Often – Always – Unsure/Not Applicable

- When thirsty, I choose to drink water.
- When thirsty, I choose a sugary drink.
- Before buying a food item, I read the nutrition label.
- When eating out with friends, I choose a healthy meal or snack (low in sugar, salt, or fat).
- When eating fruits and vegetables, I do things to make eating them more enjoyable.

Sexual Health Knowledge

- If you kiss someone with HIV, you will probably get HIV. (True / False / Unsure)
- Using condoms correctly every time you have sex helps lower your risk of HIV and STIs. (True / False / Unsure)
- The most effective way to avoid pregnancy, HIV, and STIs is to not have sex. (True / False / Unsure)
- You can have HIV or an STI and still look healthy. (True / False / Unsure)
- There is a cure for HIV. (True / False / Unsure)
- You can only get HIV if you are gay. (True / False / Unsure)
- You can lower STI risk by having one tested partner without an STI. (True / False / Unsure)
- Taking birth control pills means condoms are unnecessary. (True / False / Unsure)

- Our bodies only produce hormones during puberty. (True / False / Unsure)
- HIV and STIs can only spread when symptoms are present. (True / False / Unsure)
- Birth control can help prevent pregnancy and STIs. (True / False / Unsure)

Sexual Health Self-Efficacy

Response scale: I definitely CAN'T – I probably CAN'T – Not sure – I probably CAN – I definitely CAN

- Tell someone you don't want to have sex now.
- Decide not to make out with or kiss someone because you think it might lead to having sex.
- Tell someone you don't want to go somewhere with them because of what might happen sexually.
- Tell someone you aren't ready to have sex.
- Tell someone you do not consent to them touching you.

Sexual Health Outcome Expectations and Values

Response scale: Strongly Disagree – Disagree – Neither Agree nor Disagree – Agree – Strongly Agree

- I have at least one adult I trust who I can talk to about sex
- If I asked someone about sex, I would be nervous that they are judging me
- It is more important to me to ask questions I have about sex than it is to avoid feeling uncomfortable from asking.
- I am not afraid to ask questions about sex.
- I know the difference between rumors and facts when it comes to sex.
- I am not embarrassed to ask for help if I have a question about sex.
- I could talk to a friend about sex without being embarrassed
- If I knew the answer I would feel comfortable answering someone else's question about sex.
- I feel comfortable talking to a dating partner about what I would consent to doing sexually.

Sexual Health Peer Communication

Response scale: None – A little – Some – A lot – Unsure

- Heard from friends/students regarding
  - Ways to prevent pregnancy.
  - Sexually Transmitted Infections (STIs or STDs)
  - HIV and AIDS
  - Ways to protect yourself from STIs and HIV
  - Condoms
  - Peer pressure and sexual pressure from dating partners
  - Giving or getting consent from sexual partners

Sexual Health Parent Communication

Response scale: None – A little – Some – A lot – Unsure

- Heard from parents/guardians regarding
  - Ways to prevent pregnancy.
  - Sexually Transmitted Infections (STIs or STDs)
  - HIV and AIDS
  - Ways to protect yourself from STIs and HIV

- Condoms
- Peer pressure and sexual pressure from dating partners
- Giving or getting consent from sexual partners

Mental Health Outcome Expectations

Response scale: Strongly Disagree – Somewhat Disagree – Neither – Somewhat Agree – Strongly Agree

- If I talk about my mental health with friends or family, they will judge me.
- To be liked by others, I must keep any problems or negative feelings to myself.
- If I talk about my mental health with friends, they will say bad things about me.
- If I have problems I should not admit them to others.
- It would be a sign of weakness to show my emotions in public.
- Only people with serious mental illnesses need to talk about their mental health.
- I should not let myself feel negative emotions.
- Talking about my mental health around others makes them uncomfortable.

Mental Health Values

Response scale: Strongly Disagree – Somewhat Disagree – Neither – Somewhat Agree – Strongly Agree

- Dealing with emotions is better than ignoring them.
- I would rather ask for help than avoid judgment.
- I would rather express emotions than hide them.

Mental Health Self-Efficacy

Importance scale: Not at all Important – Slightly – Moderately – Very – Extremely

- Understanding sources of stress.
- Dealing with stressful situations well.
- Having healthy relationships with others.
- Managing conflicts with others.
- Setting limits for what you are comfortable with.
- Feeling that you belong and are accepted.

Confidence scale: Strongly Disagree – Somewhat Disagree – Neither – Somewhat Agree – Strongly Agree

- I understand my feelings when I am stressed or overwhelmed.
- I can recognize when I am angry or upset.
- I can manage my stress.
- I can calm myself down when I am mad or upset.
- I can focus on the good things in life, not just problems I am experiencing.
- I can take steps toward the future I want.
- When problems arise with my mental health or emotions, I handle them pretty well.

Self-Control

Response scale: Not at all True – A little True – Somewhat True – Pretty True – Really True

- When something frustrates me, I am able to calm down easily.

- I am able to control my temper.
- I can walk away from a situation if I am angry.
- When I have a problem, I think before I act.
- I am able to walk away when someone annoys me.
- When I am stressed, I can calm myself down.

Mental Well-Being

Response scale: Strongly Disagree – Somewhat Disagree – Neither – Somewhat Agree – Strongly Agree

- I've been feeling hopeful about the future.
- I've been feeling relaxed.
- I've been dealing with problems well.
- I've been able to make up my own mind about things.
- I've been feeling useful.
- I've been thinking clearly.
- I've been feeling good about myself.
- I've been feeling close to friends and family.

Appendix B. Data stratified by racial demographic.

Table B1. Nutrition Knowledge.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	294	3.00 (2.82, 3.18)	4.05 (3.85, 4.26)	1.05	<0.0001
Asian	1	0.00 (—)	2.00 (—)	2.00	—
American Indian or Alaska Native	3	2.33 (0.90, 3.77)	2.00 (-2.30, 6.30)	-0.33	0.8075
Hispanic or Latino	70	2.63 (2.27, 2.99)	4.29 (3.87, 4.70)	1.66	<0.0001
White	6	3.67 (1.50, 5.83)	4.17 (2.13, 6.20)	0.50	0.2956
Black or African American, Hispanic/Latino	11	2.55 (1.49, 3.60)	4.45 (3.49, 5.42)	1.91	0.0048
Other	6	2.33 (-0.38, 5.04)	4.33 (2.17, 6.50)	2.00	0.0932

Table B2. Nutrition Outcome Expectancies.

Race	n	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	294	15.66 (15.33, 15.99)	15.93 (15.57, 16.29)	0.27	0.2034
Asian	1	10.00 (—)	7.00 (—)	-3.00	—
American Indian or Alaska Native	3	14.00 (2.62, 25.38)	15.33 (3.59, 27.07)	1.33	0.6968
Hispanic or Latino	70	14.89 (14.10, 15.67)	15.00 (14.17, 15.83)	0.11	0.7926
White	6	14.67 (10.99, 18.34)	15.00 (9.37, 20.63)	0.33	0.8933
Black or African American, Hispanic/Latino	11	16.91 (14.84, 18.98)	15.82 (14.15, 17.49)	-1.09	0.2041
Other	6	13.17 (8.74, 17.59)	15.00 (9.77, 20.23)	1.83	0.1103

Table B3. Nutrition Outcome Expectancy Values.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	294	11.87 (11.61, 12.14)	11.96 (11.68, 12.25)	0.09	0.5913
Asian	1	14.00 (—)	5.00 (—)	-9.00	—
American Indian or Alaska Native	3	11.67 (6.50, 16.84)	12.00 (3.39, 20.61)	0.33	0.8845
Hispanic or Latino	70	11.53 (10.86, 12.19)	11.14 (10.43, 11.85)	-0.39	0.2991
White	6	10.17 (5.95, 14.39)	11.50 (6.46, 16.54)	1.33	0.3276

Black or African American, Hispanic/Latino	11	12.27 (10.49, 14.05)	10.82 (8.85, 12.78)	-1.45	0.0830
Other	6	9.67 (6.24, 13.09)	11.17 (7.39, 14.95)	1.50	0.0446

Table B4. Nutrition Behaviors.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	294	15.54 (15.16, 15.92)	15.39 (14.98, 15.81)	-0.15	0.5242
Asian	1	14.00 (—)	11.00 (—)	-3.00	—
American Indian or Alaska Native	3	16.00 (7.04, 24.96)	11.33 (-7.96, 30.63)	-4.67	0.2495
Hispanic or Latino	70	15.26 (14.50, 16.02)	15.39 (14.67, 16.11)	0.13	0.7282
White	6	15.83 (11.88, 19.78)	19.33 (14.80, 23.87)	3.50	0.2053
Black or African American, Hispanic/Latino	11	16.00 (14.80, 17.20)	14.82 (12.05, 17.59)	-1.18	0.3691
Other	6	17.00 (13.25, 20.75)	15.00 (11.25, 18.75)	-2.00	0.3360

Table B5. Sexual Health Knowledge.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	270	3.88 (3.58, 4.18)	6.42 (6.09, 6.75)	2.54	<0.0001
Asian	5	4.20 (1.11, 7.29)	7.20 (2.69, 11.71)	3.00	0.1249
American Indian or Alaska Native	3	3.00 (-3.57, 9.57)	4.33 (-3.65, 12.32)	1.33	0.7278

Hispanic or Latino	87	2.97 (2.47, 3.46)	6.91 (6.47, 7.35)	3.94	<0.0001
White	16	3.50 (2.10, 4.90)	8.56 (7.53, 9.59)	5.06	0.0001
Black or African American, Hispanic/Latino	9	4.33 (2.66, 6.01)	6.89 (4.91, 8.87)	2.56	0.0156
Other	5	2.20 (1.16, 3.24)	5.60 (2.48, 8.72)	3.40	0.0434

Table B6. Sexual Health Self Efficacy.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	211	18.90 (18.23, 19.56)	20.31 (19.73, 20.89)	1.42	0.0018
Asian	4	14.75 (1.73, 27.77)	20.75 (12.70, 28.80)	6.00	0.3189
American Indian or Alaska Native	2	19.00 (-31.82, 69.82)	14.50 (-106.21, 135.21)	-4.50	0.5635
Hispanic or Latino	68	18.16 (17.03, 19.29)	19.76 (18.52, 21.01)	1.60	0.0827
White	13	19.69 (17.81, 21.58)	20.23 (18.25, 22.21)	0.54	0.6934
Black or African American, Hispanic/Latino	7	18.43 (13.90, 22.95)	21.14 (18.35, 23.94)	2.71	0.2593
Other	3	16.33 (5.13, 27.53)	23.67 (20.80, 26.54)	7.33	0.1150

Table B7. Sexual Health Attitudes and Behaviors.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*

Black or African American	270	24.67 (23.96, 25.38)	31.56 (30.78, 32.34)	6.89	<0.0001
Asian	5	25.40 (21.23, 29.57)	27.80 (16.78, 38.82)	2.40	0.4375
American Indian or Alaska Native	3	27.33 (9.36, 45.30)	33.67 (19.32, 48.01)	6.33	0.0188
Hispanic or Latino	87	24.67 (23.54, 25.79)	29.62 (28.30, 30.94)	4.95	<0.0001
White	16	25.56 (22.84, 28.29)	31.31 (28.66, 33.96)	5.75	0.0001
Black or African American, Hispanic/Latino	9	26.78 (21.87, 31.68)	33.22 (29.10, 37.34)	6.44	0.0006
Other	5	20.80 (13.59, 28.01)	25.20 (17.63, 32.77)	4.40	0.0375

Table B8. Sexual Health Peer Communications.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	270	13.43 (12.74, 14.13)	16.75 (15.94, 17.56)	3.32	<0.0001
Asian	5	12.00 (5.31, 18.69)	13.40 (7.74, 19.06)	1.40	0.7250
American Indian or Alaska Native	3	15.67 (-1.24, 32.58)	15.33 (-6.37, 37.04)	-0.33	0.9098
Hispanic or Latino	87	11.15 (10.15, 12.15)	16.62 (15.18, 18.06)	5.47	<0.0001
White	16	10.38 (7.94, 12.81)	15.31 (11.94, 18.69)	4.94	0.0027
Black or African American, Hispanic/Latino	9	12.11 (8.26, 15.96)	16.78 (12.82, 20.73)	4.67	0.0565
Other	5	12.20 (4.99, 19.41)	13.40 (5.72, 21.08)	1.20	0.7899

Table B9. Sexual Health Parent Communications.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	270	12.94 (12.15, 13.74)	15.88 (15.01, 16.75)	2.94	<0.0001
Asian	5	10.80 (4.69, 16.91)	15.00 (7.81, 22.19)	4.20	0.3281
American Indian or Alaska Native	3	14.33 (1.83, 26.84)	15.33 (-0.83, 31.50)	1.00	0.7852
Hispanic or Latino	87	11.77 (10.56, 12.98)	14.63 (13.10, 16.16)	2.86	<0.0001
White	16	11.06 (8.42, 13.71)	14.25 (11.25, 17.25)	3.19	0.0507
Black or African American, Hispanic/Latino	9	12.67 (8.67, 16.66)	14.33 (10.79, 17.88)	1.67	0.2068
Other	5	10.20 (5.13, 15.27)	10.40 (4.22, 16.58)	0.20	0.9536

Table B10. Mental Health Outcome Expectations.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	257	38.00 (36.97, 39.02)	38.76 (37.83, 39.69)	0.76	0.2781
Asian	5	38.40 (32.22, 44.58)	42.40 (35.45, 49.35)	4.00	0.0716
American Indian or Alaska Native	6	41.00 (26.84, 55.16)	41.33 (35.45, 47.22)	0.33	0.9602
Hispanic or Latino	90	35.94 (34.62, 37.26)	38.26 (36.60, 39.91)	2.31	0.0394

White	4	40.00 (28.90, 51.10)	37.00 (25.02, 48.98)	-3.00	0.0805
Black or African American, Hispanic/Latino	6	41.33 (34.01, 48.65)	36.83 (25.20, 48.47)	-4.50	0.3314
Other	3	34.67 (-0.84, 70.18)	37.67 (33.87, 41.46)	3.00	0.7641

Table B11. Mental Health Values.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	257	22.91 (22.29, 23.53)	23.50 (22.90, 24.11)	0.59	0.2099
Asian	5	23.60 (19.92, 27.28)	18.40 (6.57, 30.23)	-5.20	0.3036
American Indian or Alaska Native	6	23.33 (16.01, 30.65)	17.83 (14.30, 21.37)	-5.50	0.1012
Hispanic or Latino	90	21.92 (20.94, 22.91)	23.27 (22.10, 24.43)	1.34	0.1002
White	4	20.50 (2.47, 38.53)	26.25 (19.09, 33.41)	5.75	0.1964
Black or African American, Hispanic/Latino	6	24.00 (19.26, 28.74)	20.67 (15.01, 26.32)	-3.33	0.3510
Other	3	23.67 (5.02, 42.31)	22.33 (2.41, 42.26)	-1.33	0.8579

Table B12. Mental Health Self Efficacy.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	257	25.92 (25.13, 26.71)	26.36 (25.65, 27.07)	0.44	0.4406

Asian	5	22.60 (19.36, 25.84)	22.40 (12.60, 32.20)	-0.20	0.9559
American Indian or Alaska Native	6	28.83 (22.56, 35.11)	24.33 (18.56, 30.11)	-4.50	0.1350
Hispanic or Latino	90	25.31 (24.13, 26.49)	27.02 (25.78, 28.26)	1.71	0.0560
White	4	25.00 (14.37, 35.63)	30.50 (23.81, 37.19)	5.50	0.0724
Black or African American, Hispanic/Latino	6	26.00 (22.02, 29.98)	26.83 (23.36, 30.31)	0.83	0.7494
Other	3	26.67 (12.99, 40.35)	28.00 (21.43, 34.57)	1.33	0.7735

Table B13. Mental Health Self Control.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	257	19.27 (18.52, 20.03)	20.35 (19.62, 21.09)	1.08	0.0525
Asian	5	16.20 (12.14, 20.26)	20.00 (13.26, 26.74)	3.80	0.2307
American Indian or Alaska Native	6	22.50 (14.08, 30.92)	18.33 (13.08, 23.59)	-4.17	0.3833
Hispanic or Latino	90	19.04 (17.87, 20.22)	20.87 (19.58, 22.15)	1.82	0.0347
White	4	18.25 (9.21, 27.29)	24.75 (20.00, 29.50)	6.50	0.0390

Black or African American, Hispanic/Latino	6	20.67 (16.54, 24.79)	21.17 (16.36, 25.97)	0.50	0.7905
Other	3	20.00 (12.55, 27.45)	24.00 (13.17, 34.83)	4.00	0.3001

Table B14. Mental Health Well Being.

Race	*n*	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Black or African American	257	30.35 (7.71)	30.26 (6.91)	-0.09	0.8904
Asian	5	22.60 (6.80)	25.60 (9.34)	3.00	0.4456
American Indian or Alaska Native	6	33.17 (9.68)	27.83 (9.64)	-5.33	0.4248
Hispanic or Latino	90	28.39 (6.61)	30.64 (7.73)	2.26	0.0309
White	4	25.00 (6.78)	34.75 (4.79)	9.75	0.0436
Black or African American, Hispanic/Latino	6	29.33 (7.53)	32.83 (5.27)	3.50	0.3522
Other	3	33.00 (7.00)	33.33 (6.11)	0.33	0.9658

Appendix C. Data stratified by gender identity.

Table C1. Nutrition Knowledge.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	2.80 (2.56–3.03)	3.98 (3.72–4.25)	1.19	<0.001
Boy/Man	3.03 (2.80–3.26)	4.17 (3.92–4.41)	1.13	<0.001

Table C2. Nutrition Expectancies.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	15.52 (15.10,15.94)	15.54 (15.05, ,16.04)	0.02	0.939
Boy/Man	15.46 (15.02, 15.91)	15.85 (15.41, 16.30)	0.39	0.129

Table C3. Nutrition Values.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	11.84 (11.52–12.16)	11.81 (11.42, 12.19)	-0.03	0.873
Boy/Man	11.70 (11.33–12.08)	11.69 (11.31, 12.08)	-0.01	0.963

Table C4. Nutrition Behaviors.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	15.50 (15.05, 15.95)	15.34 (14.87, 15.80)	-0.16	0.511
Boy/Man	15.66 (15.19, 16.13)	15.44 (14.87, 16.00)	-0.22	0.473

Table C5. Sexual Health Knowledge.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	3.59 (3.22, 3.96)	6.92 (6.55, 7.29)	3.33	<0.001
Boy/Man	3.67 (3.33, 4.01)	6.29 (5.92, 6.67)	2.62	<0.001

Table C6. Sexual Health Self Efficacy.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*

Girl/Woman	19.26 (18.40, 20.12)	20.37 (19.65, 21.08)	1.10	0.056
Boy/Man	18.15 (17.45, 18.85)	20.07 (19.34, 20.80)	1.92	<0.001

Table C7. Sexual Health Attitudes and Behaviors.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	24.53 (23.72, 25.35)	30.94 (30.09, 31.79)	6.41	<0.001
Boy/Man	24.96 (24.15, 25.76)	31.06 (30.09, 32.03)	6.10	<0.001

Table C8. Sexual Health Peer Communications.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	12.74 (11.97, 13.52)	17.34 (16.40, 18.27)	4.59	<0.001
Boy/Man	12.69 (11.90, 13.48)	15.79 (14.84, 16.74)	3.10	<0.001

Table C9. Sexual Health Parent Communications.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	12.57 (11.65, 13.48)	15.71 (14.68, 16.74)	3.14	<0.001
Boy/Man	12.45 (11.59, 13.32)	15.13 (14.14, 16.12)	2.68	<0.001

Table C10. Mental Health Outcome Expectations.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	38.10 (36.93, 39.27)	37.92 (36.75, 39.10)	-0.18	0.832
Boy/Man	37.15 (35.98, 38.32)	39.16 (38.12, 40.21)	2.01	0.013

Table C11. Mental Health Values.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	22.91 (22.19, 23.62)	23.30 (22.48, 24.12)	0.39	0.487
Boy/Man	22.45 (21.69, 23.21)	23.31 (22.62, 24.01)	0.86	0.133

Table C12. Mental Health Self Efficacy.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	24.66 (23.74, 25.59)	26.40 (25.52, 27.28)	1.74	0.013
Boy/Man	26.85 (25.99, 27.71)	26.66 (25.85, 27.47)	-0.18	0.761

Table C13. Mental Health Self Control.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	17.75 (16.92, 18.58)	20.29 (19.34, 21.23)	2.54	<0.001
Boy/Man	20.55 (19.70, 21.41)	20.77 (19.96, 21.58)	0.22	0.723

Table C14. Mental Health Well Being.

	Pre-Intervention μ (95% CI)	Post-Intervention μ (95% CI)	μ Difference	*p*
Girl/Woman	27.89 (26.80, 28.99)	30.21 (29.13, 31.29)	2.32	0.004
Boy/Man	31.46 (30.41, 32.51)	30.56 (29.55, 31.58)	-0.90	0.237

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